BUILD LLM

Research-Based Assessment | Training a GPT-style language model from scratch | November 2026

One-page summary

A decoder-only Transformer was trained on a 10-book Project Gutenberg text collection. Training duration and learning rate have the clearest effect on validation loss; architectural changes help more modestly over the short ten-epoch comparison window, while removing residual connections substantially impairs optimization.

Question	Observed result	Practical inference
Baseline	10 epochs: train 3.907, validation 3.937.	Loss falls steadily; extend training with validation-based stopping.
Epochs / learning rate	Best saved validation: 4.001 at epoch 37, lr=0.001.	Validation rises as training loss falls: select checkpoints, not the last epoch.
Transformer depth	7 layers: validation 4.423; 12 layers: 4.448 after 5 epochs.	More depth helps up to 7 layers in this run; 12 layers did not improve the endpoint.
Attention heads	8 heads: validation 4.462; 1 head: 4.529 after 5 epochs.	More heads give a small endpoint improvement at fixed width.
Component ablations	No residual: validation 6.796; no norm 4.202; no FFN 4.263.	Residual paths matter most here; loss does not substitute for a generation-quality test.

Key findings

1. Early stopping is essential. At lr=1e-4, validation bottoms at 3.675 at epoch 37; the 200-epoch run ends at train 0.113 / validation 6.536, strong evidence of overfitting.

2. Moderate depth is a short-run trade-off. Seven layers reached the lowest five-epoch validation loss; 12 layers were slightly worse and took longer.

3. Residual connections matter most. Without shortcuts, training plateaus near 6.95 and validation near 6.80, unlike the normalization and FFN ablations.

1. Dataset, method, and baseline

The corpus Project_Gutenberg_10_New_Releases.txt contains ten Gutenberg titles (4,425,452 raw characters); the notebook reports 4,278,693 characters after loading and preprocessing. Titles include Riders of the Wind and Lady Bell, Volume 3. Text uses GPT-2 BPE. The character stream is split 90%/10%

into training and validation partitions, then chunked into overlapping 256-token windows with stride 64.

Setting	Baseline value
Architecture	Decoder-only GPT-style Transformer; 12 blocks
Embedding width / heads	768 / 12
Context / batch	256 tokens / 16 sequences
Dropout	0.1
Optimizer	AdamW; learning rate 4e-4; weight decay 0.1
Training controls	Seed 123; gradient norm clipped at 1.0; 5 epochs
Loss logging	Mean of up to 100 training-batch losses; validation mean over first 10 batches

Baseline loss by epoch

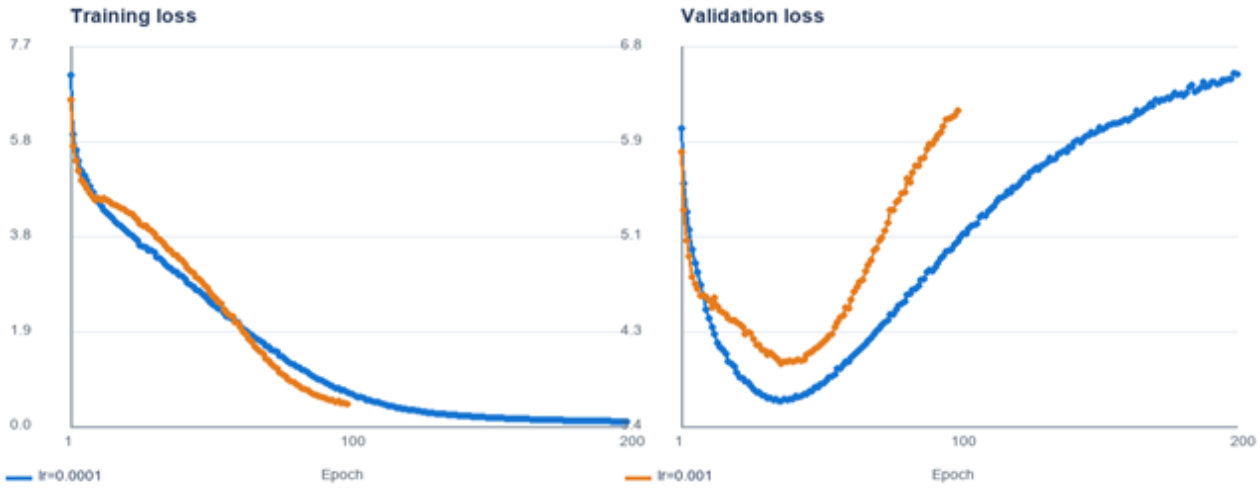


Epoch	Train loss	Validation loss
1	6.4777	5.6504
2	5.5249	5.1563
3	5.2003	4.8658
4	4.9467	4.6681
5	4.6671	4.4481

Both losses decrease across the five logged epochs, with validation loss below training loss at the endpoint. This is an early training snapshot, not evidence of convergence; longer runs show why checkpoint selection matters.

2. Experiment 1: epochs and learning rate

The saved grid contains epoch budgets of 5, 10, 20, 50, 100, and 200 for lr=1e-4, and 5, 10, 20, 50, and 100 for lr=1e-3. The figure plots the longest exported history for each learning rate; the table gives its best validation checkpoint and final logged loss.



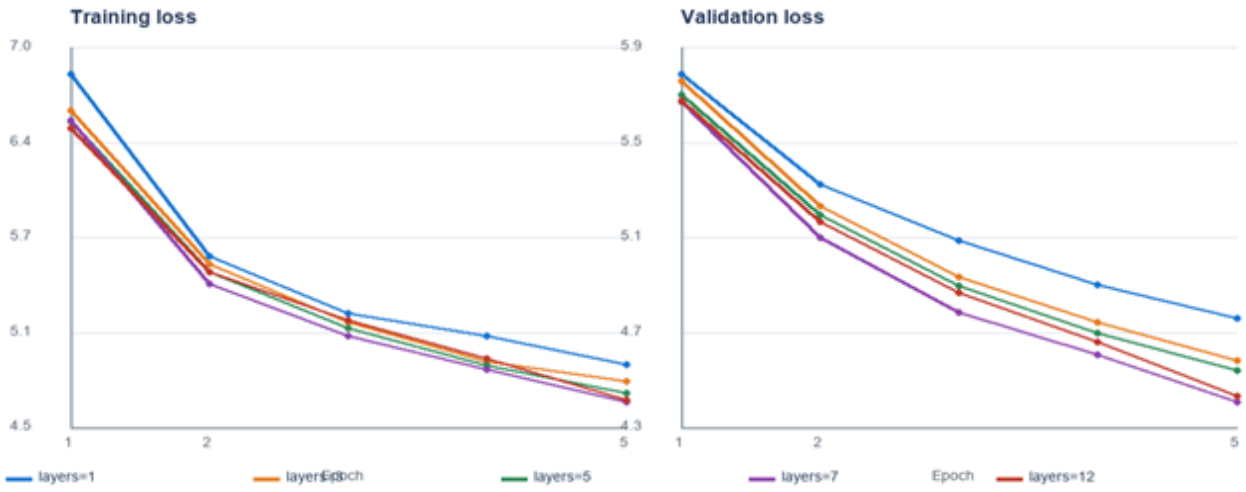
Learning rate	Longest run	Best validation (epoch)	Endpoint train / val
0.0001	200	3.675 (ep 37)	0.113 / 6.536
0.001	100	4.002 (ep 37)	0.465 / 6.219

Interpretation

The lower learning rate generalizes better in the saved runs: its best validation loss is 3.675 at epoch 37, compared with 4.002 at epoch 37 for lr=1e-3. At lr=1e-4, validation rises to 3.793 at epoch 50, 5.070 at epoch 100, and 6.536 at epoch 200 while training falls to 0.113. The lr=1e-3 history shows the same divergence (endpoint 0.465 / 6.219 at epoch 100). This supports early stopping; the smaller rate delays, but does not prevent, overfitting.

3. Experiment 2: transformer depth

Depths 1, 3, 5, 7, and 12 were trained for five epochs at learning rate 4e-4, with embedding width 768 and 12 attention heads held fixed. The panels show the complete training and validation histories.



Layers	Epoch 5 train	Epoch 5 validation	Elapsed seconds
1	4.906	4.765	94.2
3	4.797	4.591	136.2
5	4.719	4.552	177.1
7	4.658	4.423	216.0
12	4.667	4.448	318.2

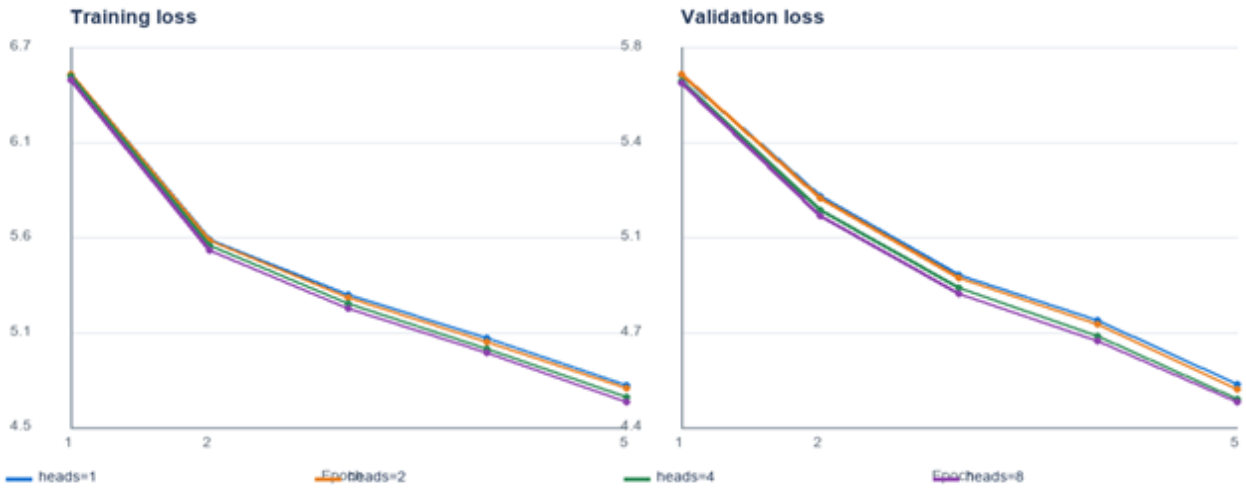
Interpretation

Five-epoch validation improves from 4.765 at one layer to 4.423 at seven layers. Twelve layers finish at 4.448, slightly worse than seven. This supports a capacity benefit through moderate depth in this short run, but not a monotonic benefit from adding layers. The seven-layer run takes about 216 seconds versus 318 seconds for 12 layers, the strongest loss/runtime trade-off among these single runs.

These are equal-epoch, single-seed comparisons, not equal-compute or repeated-seed trials. More layers add capacity and computation; loss alone cannot identify what features individual layers learned.

4. Experiment 3: attention heads

The number of heads was varied across 1, 2, 4, and 8 while holding embedding width at 768, depth at 12 layers, learning rate at 4e-4, and training length at five epochs. Increasing head count reduces each head's width from 768 to 96.



Attention heads	Head width	Epoch 5 train	Epoch 5 validation
1	768	4.782	4.529
2	384	4.765	4.509
4	192	4.718	4.472
8	96	4.690	4.462

Interpretation

Validation loss falls modestly across the tested values: 4.529 with one head, 4.509 with two, 4.472 with four, and 4.462 with eight. Multiple heads can attend through different learned projections; the slight improvement is consistent with that flexibility. The total endpoint spread is only 0.067, so the data do not support a strong claim that more heads are always better.

5. Experiment 4: component ablations

Each component was disabled throughout all 12 Transformer blocks. The no-normalization variant also replaces the final normalization with an identity operation. Ablation runs last ten epochs at learning rate 4e-4.



Configuration	Epochs	Final train	Final validation
no feedforward	10	4.308	4.263
no norm	10	3.637	4.202
no residual	10	6.945	6.796

What the ablations suggest

No normalization: optimization proceeds and reaches train 3.637 / validation 4.202, but the train-validation gap is wider near the end. Normalization controls the activation scale and can improve optimization stability; this run does not show training failure without it.

No residual connections: training stalls near 6.95 and validation near 6.80, far worse than other variants despite ten epochs. This is consistent with shortcut paths supporting signal and gradient flow through a deep stack.

No feedforward network: the model still learns (train 4.308 / validation 4.263), but lacks the block's position-wise nonlinear transformation and channel mixing. Attention alone is not enough to conclude that the FFN is dispensable.

6. Conclusions, limitations, and reproducibility

Conclusions

Within the recorded experiments, the best next-step training recipe is the smaller tested learning rate (1e-4) with checkpoint selection around the observed validation minimum, not an arbitrary long epoch budget. Seven layers are a promising efficiency candidate under the five-epoch depth study; the head-count advantage is marginal. Residual connections should be retained. Normalization

and feedforward ablation processes still learn, but their effects need matched baselines and direct generation evaluation.

Code and data appendix

The complete training and experiment code is provided in "LLM_Research _Assessment _Reference _Code _Extended _Proj _Gutenberg.ipynb". The CSV histories saved in the submission folder are the numerical source for these report's plots and tables.

Project Gutenberg corpus file: "Project_Gutenberg _10 _New _Releases.txt". The selected compilation includes ten titles